Project Overview

Ultimately, the main objective of this project is to build a web application named JobRec to help job-seekers with finding jobs that best match their confirmed skills, experience level, and location preferences through aggregated listings from company-specific recruiting sites, job APIs, and a recommendation system based on the user's professional profile.

This platform is being developed due to the fact that current job recommendation platforms, such as Linkedin, and JobRight.ai, often recommend positions which do not properly align with a user's skills, experience level, or career goals. This has resulted in many job applicants wasting time in applying to irrelevant positions, or having to manually look through job positions. JobRec overall seeks to address this problem by allowing users to upload their resumes or input their professional experience into the system, where an AI-based resume analysis tool identifies relevant skills and qualifications. Users will then be able to review and edit the extracted skill profile before recommendations are generated. Through extracting these skills, the user's experiences are then compared with aggregated job listings that have been web-scrapted from company recruiting sites and other job platforms across the internet to determine the best open job positions based on the user's skills. After parsing the data through this algorithm, it will then provide users with the most accurate and skill-relevant job postings. JobRec will also provide explanations for why these jobs have been recommended.

By improving recommendation accuracy, utilizing user preferences, and giving transparent reasoning as to why certain jobs have been recommended to the user, JobRec hopes to aid job seekers spend less time reviewing irrelevant postings. The system will also display skill gaps so users can understand whether a job is a strong match, partial match, or weak match.

Scope

In Scope:

Resume upload and skill extraction

Users will be able to upload resumes in a PDF or DOCX format, after which a skills analysis will be performed, recording their skills for later use. This is in scope because JobRec's recommendation workflow begins with the user's resume; the system must read the user's work experience, education, and skills before generating personalized recommendations.

User accounts and data management

Users will have all relevant information about themselves stored on their account, including confirmed skills, experience level, job preferences, location preferences, saved jobs, and previous searches, meaning they will not have to upload their resume again unless they want to update the information. The platform will manage their data securely. This is in scope because recommendations should persist across sessions and should be tied to the user's current profile.

Editable skill profile

After resume analysis, users will be able to review, add, remove, or edit extracted skills before recommendations are generated. This is in scope because AI-based resume parsing may miss or misclassify skills, and user correction prevents poor recommendations caused by hidden extraction errors.

Standardized skills database

JobRec will maintain a standardized skills database that maps related skill names and synonyms to consistent internal skill labels. This is in scope because resumes and job postings may describe the same skill differently, such as "JavaScript," "JS," and "ECMAScript."

Job posting database

JobRec will store collected job postings in a structured job posting database containing job title, company, location, description, required skills when available, source URL, and application link. This is in scope because the recommendation system needs stored job data to compare against user profiles.

Job data collection from external sources

The system will collect job postings from approved job APIs, company recruiting sites, and permitted public sources. This is in scope because JobRec requires a current set of job postings before it can generate useful recommendations.

Recommendation explanations

The system will display explanations for why each job was recommended, using stored user profile data and job posting data. This is in scope because JobRec's proposed value is not only recommending jobs, but also making the reasoning behind those recommendations transparent.

Skill gap detection

The system will display missing skills for recommended jobs and classify matches as strong, partial, or weak based on the number of required skills the user has. This is in scope because skill gaps help users understand why a job may or may not be a strong fit.

Skill-based job recommendations

Users will be recommended jobs that cater most to their confirmed skills, experience level, job preferences, location preferences, and the content of stored job postings. The platform will have a functionality to present the user jobs most relevant to their skills. This is in scope because recommendation generation is the central purpose of JobRec.

Basic geographic filtering

Users will be recommended jobs that cater most to their skills with a preference for locations that are close to their location. The user can either manually enter a location or allow their location to be shared. The user can also select remote, hybrid, or on-site work preferences. This is in scope because location is a major factor in whether a recommended job is relevant.

Main user interface and navigation

The platform will include the main user-facing pages needed to complete the workflow: dashboard, profile, resume upload, skill review, recommendations, saved jobs, and settings. This is in scope because users need a clear path through the product.

Authentication and user data protection

The system will require authentication for protected profile, resume, recommendation, and saved job data. This is in scope because resumes and job preferences contain sensitive personal information.

Under Evaluation:

<u>Applicant percentile or competitiveness ranking</u>

Applicant percentile ranking is under evaluation because it would require reliable data about other applicants for the same job. Without access to real applicant pools, JobRec should not claim to know how a user ranks against other candidates.

<u>Advanced employer analytics</u>

Employer analytics, such as tracking how many users viewed or matched with a posting, is under evaluation. This could be useful later, but it is not required for the first version of the job-seeker workflow.

<u>Advanced skill weighting by years of experience</u>

Advanced weighting of skills by years of experience is under evaluation. Basic experience-level matching is part of the recommendation system, but fine-grained weighting may require more user input and validation.

<u>Automated job freshness verification</u>

Automated checks for whether a job is still open are under evaluation. This could improve job quality, but it may require repeated API checks or source-specific logic.

Out of Scope:

<u>Social media capabilities</u>

JobRec will not be LinkedIn; there will be no capacity for networking features such as feeds, messaging, endorsements, posts, or connections. The platform is solely focused on providing good and transparent job recommendations rather than allowing users to interact with each other actively.

<u>Fully automated job applications</u>

JobRec will not automatically apply to jobs on behalf of users. It will provide recommendations and application links, but users will complete applications externally.

Full recruiter or applicant tracking system

JobRec will not replace an applicant tracking system. Employer workflows such as reviewing candidates, sending interview invitations, and managing hiring pipelines are outside the current project scope.

Guaranteed applicant percentile ranking

JobRec will not guarantee percentile rankings against real applicants unless reliable applicant pool data becomes available.

Resume template builder

JobRec will not provide a full resume builder or resume formatting tool. The system may analyze uploaded resumes, but creating or formatting resumes is outside the current scope.

Assumptions

We are assuming that the users are job seekers who already have a resume and already have some knowledge about how to apply for a job.

Users have an existing resume in PDF or DOCX format or are willing to manually enter equivalent profile information.

Selected job APIs, company recruiting pages, or permitted public sources provide enough information to extract job title, company, location, description, required skills, and application links.

Users are willing to review and correct extracted skills before relying on recommendation results.

A standardized skills database can cover the most common skills needed for the initial target users, especially software, engineering, business, and technical roles.

The first version of JobRec will be evaluated using representative test resumes and job postings rather than a full production applicant pool.

Constraints

External API limits: Job APIs may impose rate limits, request quotas, authentication requirements, pricing limits, pagination restrictions, or data access limits. These constraints can reduce how often JobRec refreshes job postings and how many postings can be collected during a given time window.

External source availability: Company career pages and job APIs may change formats, become unavailable, block scraping, or remove postings without notice. This can affect job coverage and data freshness.

Incomplete job descriptions: Some job postings may omit required skills, experience level, salary, work authorization details, or location information. Missing fields can reduce recommendation quality.

LLM cost and response variability: AI-based extraction and explanation generation may have cost, latency, and consistency constraints. The system must limit AI usage to high-value steps and ground outputs in stored data.

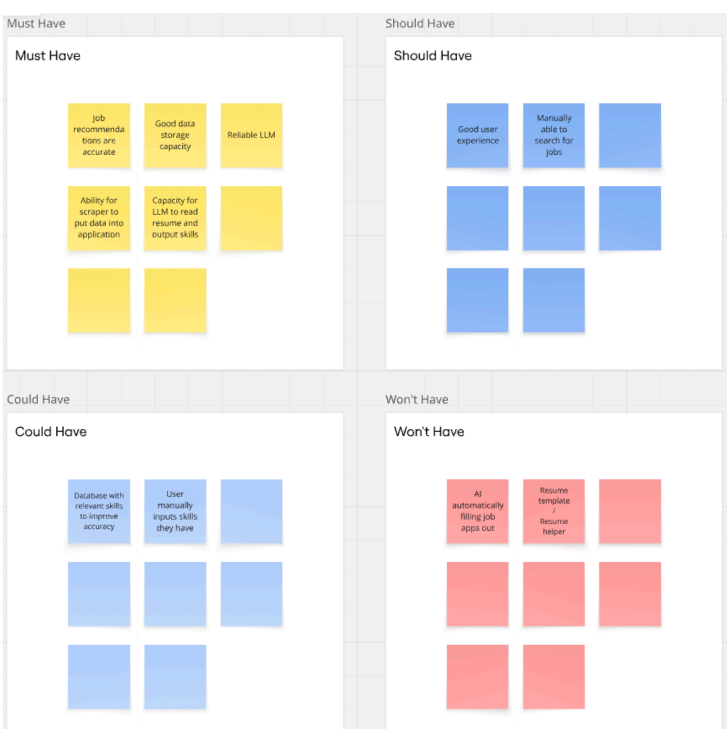
Privacy and security requirements: Because resumes and user profiles may contain personal information, authentication, authorization, and secure storage requirements constrain the system design.

Project timeline: The team must prioritize the core workflow of resume upload, skill confirmation, job collection, recommendations, and explanations before advanced features such as employer analytics or applicant ranking.

Requirements Elicitation & Prioritization

Moscow Method Reflection

Team 1 conducted a MoSCoW method brainstorming session on Thursday, March 5th. During brainstorming, the team first listed candidate capabilities before sorting them into final MoSCoW categories. Initial candidate requirements included resume upload, PDF/DOCX parsing, skill extraction, editable skill review, user account creation, authentication, profile persistence, job data collection, job posting storage, job recommendation generation, recommendation explanations, location filtering, skill gap detection, saved jobs, employer-submitted postings, job link validation, applicant percentile ranking, automated job applications, social networking features, and resume template generation. We reflected on our stakeholders and discussed what we must have, should have, could have, and won't have in our project. We found discussing each MoSCoW category useful because it forced the team to separate the minimum viable JobRec workflow from helpful but nonessential features. The must-have category focused on the core user flow: creating an account, uploading a resume, reviewing extracted skills, generating recommendations, viewing explanations, and saving profile data. The should-have and could-have categories captured useful enhancements such as standardized skills, employer-submitted postings, and advanced skill weighting. The team found "acting as stakeholders" less useful at this stage because we did not yet have enough direct interview data from recruiters, employers, or job seekers to confidently represent their priorities. Instead, peer feedback and instructor feedback were more useful for identifying missing requirements and scope gaps. Below is a screenshot of our moscow method matrix.



Updated MoSCoW Prioritization Summary

Must Have:
- User account creation and authentication
- Resume upload
- Resume text processing
- Skill extraction
- Editable skill profile
- Job posting database
- Job data collection
- Ranked job recommendations
- Recommendation explanations
- Skill gap detection
- Job posting link display
- User profile persistence
- Main UI/navigation
- User data protection

Should Have:
- Standardized skills database
- Location preference matching
- Search/profile refinement
- Employer job posting submission
- Job link validity checking
- Data storage performance testing
- Usability/accessibility testing

Could Have:
- Advanced skill weighting by years of experience
- Automated job freshness verification
- Employer analytics
- Saved comparison view between multiple jobs
- Advanced recommendation tuning controls

Won’t Have:
- Social networking feed
- Messaging between users
- Fully automated job applications
- Full recruiter/ATS workflow
- Guaranteed applicant percentile ranking
- Resume template builder

<u>Software Requirements</u>

Functional Requirements:

1. FR-001 - Generate Ranked Job Recommendations
    a. Statement:
        i. When a user has completed and confirmed their profile, which includes adding their resume, skills, experience level, etc., the system shall generate and provide a ranked list of job postings whose required skills and qualificants are a close match to the user's confirmed skills, experience level, location preferences, and job preferences.
    b. Rationale:
        i. The main goal of this platform is to provide users with a recommendation list so users do not have to manually search through large numbers of job postings.
    c. Test Method:
        i. Create 25 representative user profiles and a test database of at least 500 job postings. For each user profile, request recommendations and verify that the system returns a ranked list of at least 10 job postings when matching postings exist.
    d. Supporting Context:
        i. User profile: Contains stored information of a sample user, which includes skills, education level, work experience, and user's job preferences (What types of jobs they are looking for).
        ii. Ranked list: Actual jobs recommended based on the user's skills, ordered by their overall skills match and relevance to the user's background.
    e. Tracing Information:
        i. SYS-001 Accurate Job Recommendations
        ii. SYS-005 Persistent and Accurate Data Management
    f. Priority:
        i. Must Have

2. FR-002 - Display Clickable Job Posting Links
    a. Statement:
        i. When a user is selecting a job listing from their recommendation list, the system should provide a link that will redirect the user to the original job application page that is associated with the job posting.
    b. Rationale:
        i. Users need a direct path from recommendation review to the actual job application source.
    c. Test Method:
        i. Randomly sample 100 stored job postings. When opening the application link, ensure that 99% of the links are not broken, are displayed, and direct the user to the actual job posting and application page.
    d. Supporting Context:
        i. Application link: the URL is directing users to the employer's job application page.

   ii. Correct page: the page that contains the job listing matches the description on the JobRec platform.

e. Tracing Information:
   i. SYS-002 Accurate Job Scraping
   ii. SYS-006 User Interface

f. Priority:
   i. Must have

3. FR-003 - Collect and Store Job Posting Data
   a. Statement:
      i. When the system is collecting job postings from job sources across the web, the system shall extract and collect the job title, company name, job location, job description, and application link in the job posting database.
   b. Rationale:
      i. A structured database of jobs is required to make sure that there are reliable recommendations and relevant job information for users.
   c. Test Method:
      i. Run the job collection process on at least three approved sources. Verify that at least 300 total job posting records are created or updated in the database and that at least 95% contain the required fields when those fields are available from the source.
   d. Supporting Context:
      i. Supporting job sources: websites/APIs that have been used by the platform to collect job postings.
      ii. Extraction: retrieval by algorithm of job information from source.
      iii. Job posting database: persistent database table or collection containing normalized job posting records.
   e. Tracing Information:
      i. SYS-002 Accurate Job Scraping
      ii. SYS-005 Persistent and Accurate Data Management
   f. Priority:
      i. Must have

4. FR-004 - Generate Grounded Recommendation Explanations
   a. Statement:
      i. When the system job recommendations, the system shall generate explanations by utilizing only verified data from the user profile and stored job posting information.
   b. Rationale:
      i. This prevents misleading information and increases user trust; each explanation must be traceable to specific stored fields.
   c. Test Method:
      i. Generate 50 explanations for recommended jobs. Ensure at least 95% of the explanation statements match the actual user’s skills that are stored in the database.

d. Supporting Context:
    i. Grounded explanation: recommendation explanation that can be traced to stored data fields.
    ii. Verified data: fields that are within a platform database, such as skills, experience, and job requirements.
e. Tracing Information:
    i. SYS-001 Accurate Job Recommendations
f. Priority:
    i. Must Have

5. FR-005 - Location Preference Matching
    a. Statement:
        i. When a user specifies location preferences, such as certain geographic regions, or remote/hybrid/on-site work, the system will prioritize job recommendations which accurately match the specific location preferences.
    b. Rationale:
        i. Location constraints are a large factor in job selection for users. When determining accurate job recommendations, it is important to consider location as well.
    c. Test Method:
        i. Create 50 test profiles that have listed specific location preferences. Ensure that at least 80% of the job recommendations to each user accurately match the location preferences of the sample user.
    d. Supporting Context:
        i. Location preference: geographic areas, states, cities, or remote work options that have been selected by the user.
    e. Tracing Information:
        i. SYS-001 Accurate Job Recommendations
    f. Priority:
        i. Must Have

6. FR-006 - Skill Gap Detection
    a. Statement:
        i. When the system determines a recommended job requiring skills that are not present in the user profile, the system shall identify and display the missing skills that are required for the job that the user does not have and categorize the match as strong (80% or more), partial (50%-79%), or low (less than 50%) based on the size of the gap.
    b. Rationale:
        i. Skill gap detection supports recommendation transparency. It also helps define how much mismatch is tolerable. A job with one or two missing preferred skills may still be relevant, while a job missing many required skills should be marked as a weaker match. Insights as to what skill gaps the user has for the job helps the user, for the future, understand the

qualifications that they need in order to become more eligible for the roles that they want.

c. Test Method:
    i. Create 50 recommended job/profile pairs with known required skills and known user skills. Verify that the system displays the correct missing skills and assigns the correct match category for at least 90% of test pairs.
d. Supporting Context:
    i. Skill gap: the difference between skills required by a job posting and the skills that have been listed by the user in their profile.
e. Tracing Information:
    i. SYS-001 Accurate Job Recommendations
    ii. SYS-003 Skill Gap Detection
f. Priority:
    i. Should Have

7. FR-007 - Editable Skill Profile
    a. Statement:
        i. When the system extracts skills from a user's resume, the system shall allow the user to review, edit, add, or remove skills before recommendations are generated.
    b. Rationale:
        i. AI-based extraction may miss or incorrectly identify skills. User review prevents poor recommendations caused by hidden parsing errors.
    c. Test Method:
        i. Create 20 test user profiles with uploaded resumes. Verify that each user can add, remove, and edit extracted skills. Refresh the page and confirm that all edits persist. Regenerate recommendations and verify that the updated skills are used.
    d. Supporting Context:
        i. Editable skill profile: user-controlled list of confirmed skills.
    e. Tracing Information:
        i. SYS-001 Accurate Job Recommendations
        ii. SYS-005 Persistent and Accurate Data Management
        iii. SYS-006 User Interface
    f. Priority:
        i. Must Have

8. FR-008 - Maintain Standardized Skills Database
    a. Statement:
        i. The system shall maintain a standardized skills database that maps skill synonyms and variations to consistent internal skill names.
    b. Rationale:
        i. Standardized skills improve matching consistency between resumes and job postings. For example, "JS" and "JavaScript" should map to the same internal skill.

c. Test Method:
    i. Create a database of at least 500 standardized skills and synonyms. Run 50 test skill queries using abbreviations, alternate spellings, or related phrases. Verify that the correct standardized skill appears in the top 5 results at least 90% of the time.
d. Supporting Context:
    i. Standardized skill: normalized skill name used internally by JobRec.
e. Tracing Information:
    i. SYS-005 Persistent and Accurate Data Management
    ii. SYS-003 Skill Gap Detection
f. Priority:
    i. Should Have

9. FR-009 - Process Uploaded Resume Files
    a. Statement:
        i. When a user uploads a PDF or DOCX resume, the system shall process the file and convert it into readable text for skill extraction.
    b. Rationale:
        i. Resume upload is the entry point for the recommendation workflow. The system must convert resumes into usable text before extracting skills.
    c. Test Method:
        i. Create 20 sample resumes and export each as PDF and DOCX. Upload all 40 files. Verify that the system extracts readable text from at least 90% of files and rejects unsupported/corrupted files with a clear error message.
    d. Supporting Context:
        i. Resume parser: component that extracts readable text from uploaded resume files.
    e. Tracing Information:
        i. SYS-004 Resume Upload and Analysis Performance
    f. Priority:
        i. Must Have

10. FR-010 - Provide Main UI Navigation
    a. Statement:
        i. When a user is logged in, the system shall provide navigation between the dashboard, profile page, resume upload page, skill review page, recommendations page, saved jobs page, and settings page.
    b. Rationale:
        i. Users need a clear interface to complete the JobRec workflow. Without navigation, the system may have backend functionality but no usable path through the product.
    c. Test Method:
        i. Create a test account and verify that each main page is reachable from the navigation menu. Confirm that no required workflow step is hidden or inaccessible.

d. Supporting Context:
   i. UI: website for users to manage their user profile and view recommendations.
e. Tracing Information:
   i. SYS-006 User Interface
f. Priority:
   i. Must Have

11. FR-011 - Authenticate Users
   a. Statement:
      i. When a user creates an account or logs in, the system shall authenticate the user before allowing access to protected profile, resume, recommendation, or saved job data.
   b. Rationale:
      i. JobRec stores sensitive resume and profile data, so protected features must require authentication.
   c. Test Method:
      i. Attempt to access five protected pages while logged out. Verify that each request redirects to login or returns an unauthorized response. Then log in and verify that the authenticated user can access their own protected pages.
   d. Supporting Context:
      i. User Account: stores user information including resume, skills, education, experience, and more.
   e. Tracing Information:
      i. SYS-006 User Interface
      ii. SYS-005 Persistent and Accurate Data Management
   f. Priority:
      i. Must Have

12. FR-012 - Create User Profile
   a. Statement:
      i. When a new user signs up, the system shall create a user profile that can store resume information, confirmed skills, preferences, saved jobs, and previous searches.
   b. Rationale:
      i. A persistent profile is required for personalized recommendations and repeated use of the platform.
   c. Test Method:
      i. Create 20 new user accounts. Verify that each account receives a profile record and that the profile can store skills, preferences, and saved job data.
   d. Supporting Context:
      i. User Account: stores user information including resume, skills, education, experience, and more.
   e. Tracing Information:

        i. SYS-006 User Interface
        ii. SYS-005 Persistent and Accurate Data Management
    f. Priority:
        i. Must Have

13. FR-013 - Refine Search and Profile Preferences
    a. Statement:
        i. When a user updates their skills, experience level, role preference, location preference, or work-mode preference, the system shall use the updated information in future recommendation requests.
    b. Rationale:
        i. Users may change goals or correct profile data over time. Recommendations must reflect the latest user-confirmed information.
    c. Test Method:
        i. Create 20 profiles, generate initial recommendations, update at least one preference or skill for each profile, and regenerate recommendations. Verify that updated fields are used in the new recommendation request.
    d. Supporting Context:
        i. User profile: Contains stored information of a sample user, which includes skills, education level, work experience, and user's job preferences (What types of jobs they are looking for).
        ii. Editable skill profile: user-controlled list of confirmed skills.
    e. Tracing Information:
        i. SYS-006 User Interface
        ii. SYS-001 Accurate Job Recommendations
    f. Priority:
        i. Must Have

14. FR-014 - Persist Profiles and Previous Searches
    a. Statement:
        i. When a user saves profile information, confirmed skills, job preferences, saved jobs, or previous searches, the system shall persist that data and reload it in future sessions.
    b. Rationale:
        i. Persistence is a functional requirement because JobRec users should not need to recreate their profile each time they log in.
    c. Test Method:
        i. Create 20 user profiles, save profile fields and searches, log out, log back in, and verify that all saved profile fields and previous searches are restored.
    d. Supporting Context:
        i. User profile: Contains stored information of a sample user, which includes skills, education level, work experience, and user's job preferences (What types of jobs they are looking for).
    e. Tracing Information:

    i. SYS-006 User Interface
    ii. SYS-005 Persistent and Accurate Data Management
f. Priority:
    i. Must Have

15. FR-015 - Save Recommended Jobs
    a. Statement:
        i. When a user views a recommended job, the system shall allow the user to save the job to their account for later review.
    b. Rationale:
        i. Users often compare jobs over multiple sessions. Saving jobs improves practical usability and supports the job-search workflow.
    c. Test Method:
        i. For 20 test users, save 5 recommended jobs each. Log out and log back in, then verify that all saved jobs are still associated with the correct user account.
    d. Supporting Context:
        i. User profile: Contains stored information of a sample user, which includes skills, education level, work experience, and user's job preferences (What types of jobs they are looking for).
    e. Tracing Information:
        i. SYS-001 Accurate Job Recommendations
        ii. SYS-005 Persistent and Accurate Data Management
    f. Priority:
        i. Should Have

Non-functional Requirements:

1. NFR-001 - Webpage Load Performance
    a. Statement:
        i. Under average system load (up to 25 concurrent active users, a database containing up to 10,000 job postings, and normal recommendation/search traffic without artificial stress testing), 95% of main user-facing pages shall display visible content within 2 seconds.
    b. Rationale:
        i. Page load time affects usability.
    c. Test Method:
        i. Conduct 500 page load tests under average/expected system load. Ensure that 95% of the time, the page is loaded and ready within 2 seconds.
    d. Supporting Context:
        i. Visible page content: the website, subpages, resume upload section, etc. Everything that is displayed to the user.
    e. Tracing Information:
        i. SYS-006 User Interface
    f. Priority:
        i. Must Have

2. NFR-002 - User Interface (UI) Usability
    a. Statement:
        i. When a user interacts with the platform interface, the system shall provide a visually consistent, aesthetic, and user-friendly design that has an average usability score that is >= 80 on the System Usability Scale (SUS).
    b. Rationale:
        i. An aesthetic, and easy-to-use interface improves user engagement, and platform understanding.
    c. Test Method:
        i. Conduct usability testing with 10 users. Collect the survey responses, and verify that the SUS score is on average, greater than or equal to 80.
    d. Supporting Context:
        i. System Usability Scale (SUS): Standard usability measurement scored from 0-100 to determine how user-friendly a platform is.
    e. Tracing Information:
        i. SYS-006 User Interface
    f. Priority:
        i. Must Have

3. NFR-003 - User Data Protection
    a. Statement:
        i. When a user uses an account to access their skills or job recommendations, or their employment history, full name, address, or

other PII, JobRec shall verify that they are the owner of the account before providing any information.

b. Rationale:
    i. Because JobRec directly interfaces with the user's employee life through job histories, current employment, and future employment opportunities, it is necessary for it to collect PII on them to find accurate and relevant job listings. Since this information can be used by bad actors to impersonate them or perform doxxing attacks, it is critical that this information only be accessible to the user it belongs to.
c. Test Method:
    i. We will use the OWASP ZAP web app scanner to ensure that authentication flows are secure and prevent unwanted access. We will also use the compliance checker tool Compliance Masonry and manual checklist verification to determine whether the website is compliant with NIST SP 800-63B-4 guidelines for authentication.
d. Supporting Context:
    i. PII refers to Personally-Identifying Information, which can be used to link a digital identity to a physical one. NIST SP 800-63B-4 are guidelines for a webserver to establish the identity of users and verify that they own the identity they are trying to authenticate with.
e. Tracing Information:
    i. SYS-006 User Interface
f. Priority:
    i. Must Have

4. NFR-004 - Data Storage Performance
    a. Statement:
        i. While JobRec has a large number of stored job applications and skills, when a user requests job recommendations, JobRec shall continue to provide them in a performant way.
    b. Rationale:
        i. Though we are only expecting double-digit users, the system will still be scraping many thousands of job applications, and a database of skill names for the LLM to use (if that is how we implement consistent skill name usage) will be considerably large. As such, it is imperative that JobRec maintains performance in this scenario and can efficiently filter through that large repository of job listings to find ones relevant to the user.
    c. Test Method:
        i. The job database will be filled with 10,000 sample job listings. Then, a new user account will be created with a sample resume, and the user will query for job recommendations. The time for the response to be generated will then be recorded. This process will be repeated over 5 trials to minimize outliers. The resulting average time should remain under 10 seconds.
        ii. We will also measure the performance of job recommendation querying under scenarios where the database contains 10, 100, 1,000, and 10,000

jobs. The time complexity relative to database entries should then be experimentally determined from this data, and it should be at most linear.

d. Supporting Context:
   i. Recommendation query: backend operation that compares a user profile against stored job postings and returns ranked results.
e. Tracing Information:
   i. SYS-006 User Interface
f. Priority:
   i. Should Have

5. NFR-005 - Recommendation Match Quality
   a. Statement:
      i. For representative test profiles, at least 80% of the top 10 recommended jobs shall be judged relevant based on confirmed skills, experience level, and user preferences.
   b. Rationale:
      i. Recommendation quality is a non-functional quality attribute, not the functional behavior itself.
   c. Test Method:
      i. Create 25 representative user profiles and a test database of at least 500 job postings. Generate the top 10 recommendations for each profile. A recommendation is relevant if it matches the user's confirmed skills, experience level, and preferences according to a defined evaluation rubric. At least 80% of top-10 recommendations must be relevant.
   d. Supporting Context:
      i. N/A
   e. Tracing Information:
      i. SYS-001 Accurate Job Recommendations
   f. Priority:
      i. Must Have

6. NFR-006 - Job Link Validity
   a. Statement:
      i. At least 95% of stored job application links shown to users shall resolve to an active webpage and match the stored company or job context.
   b. Rationale:
      i. Displaying a clickable link is functional, but link correctness is a quality attribute.
   c. Test Method:
      i. Use an automated link checker to test 100 stored job application URLs. For links that return valid HTTP responses, verify that the target page title or content contains the expected company or job title where available.
   d. Supporting Context:
      i. N/A
   e. Tracing Information:

        i. SYS-002 Accurate Job Scraping
    f. Priority:
        i. Must Have

7. NFR-006 - Job Data Extraction Quality
    a. Statement:
        i. For supported job sources, at least 95% of collected postings shall contain correctly populated title, company, location, description, and application link fields when those fields are present in the source.
    b. Rationale:
        i. Collecting job data is functional, but extraction correctness is a data quality attribute.
    c. Test Method:
        i. Run the collector on approved sources and use automated field validation to check missing fields, malformed URLs, duplicate postings, and inconsistent company/title fields. Manually review a smaller audit sample of 25 postings to confirm automated checks.
    d. Supporting Context:
        i. N/A
    e. Tracing Information:
        i. SYS-001 Accurate Job Scraping
    f. Priority:
        i. Must Have

Deliverables

Deliverable: A managed job collection pipeline using AWS EventBridge Scheduler or an equivalent modern scheduling service to periodically collect, normalize, validate, and store job postings from approved sources.

- **Requirements:** FR-003 Accurate Job Data Scraping, FR-001 Accurate Job Recommendations, FR-002 Valid Job Applications Links/Redirection, FR-005 Location Preference Matching, NFR-004 Data Storage

Deliverable: A deployed web application that allows users to create accounts, upload resumes, review/edit extracted skills, set preferences, view ranked job recommendations, inspect recommendation explanations, view skill gaps, and save jobs.

- **Requirements:** FR-001 Accurate Job Recommendations, FR-002 Valid Job Applications Links/Redirection, FR-005 Location Preference Matching, FR-006 Skill Gap Detection, NFR-001 Fast Webpage Performance, NFR-002 Modern and Visually Appealing User Interface, NFR-004 Data Storage

Deliverable: A backend API and database system that stores user profiles, resumes, confirmed skills, standardized skill mappings, job postings, previous searches, and saved jobs.

- **Requirements:** FR-003, FR-008, FR-012, FR-013, FR-015, FR-016, NFR-003, NFR-004

Deliverable: A resume processing module that accepts PDF/DOCX uploads, extracts readable text, identifies candidate skills, maps them to standardized skills, and presents them for user review.

- **Requirements:** FR-007, FR-008, FR-009, FR-013, FR-014

Deliverable: A system accuracy report measuring the following on a sample of 100 job listings and 50 user resumes: the number of correctly-extracted skills by the system, the number of hallucinated skills, the percentage of skill gaps correctly identified in each match, the percentage of functioning links to original listings, the human-verified and qualitative accuracy of matched job listings, and the percentage of job recommendations that match user location preferences. Extra listings may also be included in this sample for certain tests, but each must use a minimum of 100.

- **Requirements:** FR-001 Accurate Job Recommendations, FR-002 Valid Job Applications Links/Redirection, FR-003 Accurate Job Data Scraping, FR-004 ChatGPT Hallucination Minimization, FR-005 Location Preference Matching, FR-006 Skill Gap Detection

Deliverable: A security and compliance audit report demonstrating that the authentication methods and data storage behaviors of the system conform to the NIST SP 800-63B-4 Digital Identity Guidelines. The report will include the results of testing authentication flows with OWASP ZAP, manual checklist verification, and gap analysis via Compliance Masonry.

- **Requirements:** NFR-003 user data protection

Deliverable: A report containing performance test results for page loading, recommendation query latency, and database-size scaling.

- **Requirements:** NFR-004 Data Storage

Deliverable: A system usability report containing a usability test with 10 users, 500 page load tests and their load times, and compliance with WCAG 2.2. The latter will be done using the WAVE tool pointed at our production server.

- **Requirements:** NFR-001 Fast Webpage Performance, NFR-002 Modern and Visually Appealing User Interface

Standards

Web Content Accessibility Guidelines (WCAG) 2.2: https://www.w3.org/TR/WCAG22/

- This standard is applicable because our project is a web-based system that job-seekers will use to search jobs, upload information, and review recommendations; hence, the interface needs to be accessible to users with visual, motor, and cognitive disabilities. It will impact our design and code by requiring features like keyboard-navigable UI flows, sufficient color contrast, form labels, accessible error messages, readable font sizes, and more. We will ensure compliance with the standard by running automated accessibility checks during development and manually testing workflows with keyboard-only navigation and screen readers.

Digital Identity Guidelines: Authentication and Authenticator Management: https://csrc.nist.gov/pubs/sp/800/63/b//final

- This standard is applicable because our system has user accounts, logins, and will store sensitive profile data such as resumes, skills, job preferences, visa status, locations, and more. As a result, authentication needs to follow a recognized industry standard; this standard provides information related to defining technical requirements for authenticator assurance levels and focuses on authentication over networks. It will impact our design and code by shaping password handling, MFA support, session/authenticator lifecycle rules, and the way we implement signup, login, password reset, and account recovery. We will ensure compliance with the standard by using Auth0, enforcing strong password and session policies, and documenting authentication requirements in our system design and test cases.

OpenID Connect 1.0: https://openid.net/specs/openid-connect-core-1_0.html

- This standard is applicable because since we're using a third-party identity services tool (Auth0) for secure sign-in, OpenID Connect is the standard identity layer built on top of OAuth 2.0, which Auth0 uses, helping to verify end-user identity and obtaining profile claims. It will impact our design and code by determining how the frontend and backend handle login redirects, ID tokens, user claims, session validation, and integration with Auth0. We will ensure compliance with the standard by using a standards-compliant OpenID Connect provider, validating tokens on the backend, and testing authentication flows against the OpenID connect specification instead of using custom login logic.